

Research Report on Common Network Security Threats

Submitted By: Naman

GitHub Repository: OIBSIP-CyberSecurity

Organization: Oasis Infobyte

Internship Domain: Cyber Security Internship

Task Number: Task 4

Table of Contents

- Introduction
- What is Network Security?
- Common Network Security Threats
- Denial of Service (DoS)
- Distributed Denial of Service (DDoS)
- Man-in-the-Middle (MITM)
- IP Spoofing
- DNS Spoofing
- Malware
- Ransomware
- Real-World Case Studies
- Prevention and Mitigation
- Best Practices
- Conclusion
- References

1. Introduction

The rapid growth of the internet and digital technologies has transformed the way individuals and organizations communicate, store information, and conduct business. While these advancements provide numerous benefits, they also expose networks to a wide range of cyber threats. Attackers continuously search for vulnerabilities to steal confidential information, disrupt services, or gain unauthorized access to systems.

Network security is the process of protecting computer networks, devices, and data from cyberattacks. It combines security technologies, policies, and best practices to ensure that information remains secure and accessible only to authorized users.

This report discusses common network security threats, explains how they work, highlights their impact through real-world examples, and presents effective prevention techniques.

2. What is Network Security?

Network security refers to the practice of protecting a computer network from unauthorized access, cyberattacks, data breaches, and misuse. It involves the implementation of hardware devices, software applications, firewalls, antivirus solutions, intrusion detection systems, encryption techniques, and security policies.

The primary goals of network security are based on the CIA Triad:

Confidentiality – Ensuring that sensitive information is accessible only to authorized users.

Integrity – Protecting data from unauthorized modification or corruption.

Availability – Ensuring that systems and services remain available whenever required.

A secure network minimizes security risks, protects sensitive information, and ensures smooth business operations.

3. Common Network Security Threats

3.1 Denial of Service (DoS) Attack

A Denial of Service (DoS) attack is a cyberattack in which an attacker attempts to make a computer system, website, or network unavailable to legitimate users by overwhelming it with excessive traffic or malicious requests. The main objective of a DoS attack is to exhaust system resources such as CPU, memory, bandwidth, or storage so that genuine users cannot access the service.

How a DoS Attack Works :

- The attacker targets a specific server or network.
- A large number of fake requests are sent to the target.
- The server becomes overloaded while processing these requests.
- Legitimate users experience slow performance or complete service unavailability.

Impact of DoS Attacks

- Website or application downtime.
- Financial losses due to interrupted business operations.
- Loss of customer trust and reputation.
- Reduced productivity within organizations.

Prevention

- Configure firewalls and Intrusion Prevention Systems (IPS).
- Implement rate limiting to control excessive traffic.
- Use Content Delivery Networks (CDNs).
- Monitor network traffic continuously.
- Keep systems updated with the latest security patches.

3.2 Distributed Denial of Service (DDoS) Attack

A Distributed Denial of Service (DDoS) attack is an advanced version of a DoS attack. Instead of using a single computer, the attacker controls thousands of infected devices (known as a botnet) to send massive amounts of traffic simultaneously to the target server. Because requests originate from multiple locations, DDoS attacks are much more difficult to detect and block.

How a DDoS Attack Works :

- Malware infects thousands of computers or IoT devices.
- These infected devices become part of a botnet.
- The attacker instructs the botnet to send simultaneous requests.
- The target server becomes overloaded and stops responding.

Impact of DDoS Attacks :

- Complete network or website outage.

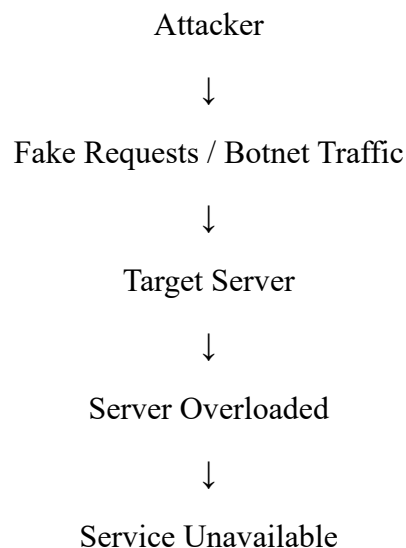
- Increased operational costs.
- Service disruption affecting customers.
- Significant business and reputational damage.

Real-World Example

In October 2016, the Dyn DNS DDoS Attack disrupted major online services including Twitter, Netflix, Reddit, GitHub, and Spotify. Attackers used the Mirai Botnet, which infected thousands of IoT devices such as cameras and routers. This incident demonstrated how insecure connected devices can be exploited to launch large-scale cyberattacks.

Prevention :

- Deploy DDoS protection services.
- Use cloud-based traffic filtering.
- Maintain redundant network infrastructure.
- Monitor unusual traffic patterns.
- Implement strong security measures for IoT devices



3.3 Man-in-the-Middle (MITM) Attack

A Man-in-the-Middle (MITM) attack is a cyberattack in which an attacker secretly intercepts and sometimes alters the communication between two parties without their knowledge. Both parties believe they are communicating directly with each other, while the attacker is actually monitoring or manipulating the exchanged data.

MITM attacks commonly occur on unsecured public Wi-Fi networks, compromised routers, or through malicious software. Attackers use these attacks to steal usernames, passwords, banking information, confidential documents, and other sensitive data.

How a MITM Attack Works :

- The attacker places themselves between the sender and the receiver.
- Data sent by the sender first reaches the attacker.
- The attacker can read, modify, or record the data.
- The attacker forwards the data to the receiver without raising suspicion.

Impact of MITM Attacks :

- Theft of login credentials.
- Financial fraud.
- Identity theft.
- Leakage of confidential business information.
- Unauthorized access to sensitive systems.

Real-World Example

Many attackers create fake public Wi-Fi hotspots in places such as airports, cafes, and hotels. When users connect to these networks, attackers capture sensitive information such as passwords, emails, and banking details.

Prevention :

- Always use HTTPS websites.
- Avoid using public Wi-Fi for sensitive activities.
- Enable Virtual Private Network (VPN).
- Use Multi-Factor Authentication (MFA).
- Keep devices and browsers updated.

3.4 IP Spoofing

IP Spoofing is a technique in which an attacker disguises their real IP address by replacing it with a fake or trusted IP address. This makes malicious traffic appear legitimate and helps attackers bypass certain security mechanisms.

IP spoofing is commonly used in DDoS attacks, session hijacking, and network reconnaissance.

How IP Spoofing Works :

- The attacker modifies the source IP address of data packets.
- The target system believes the packets originate from a trusted source.
- The attacker gains unauthorized access or launches further attacks.

Impact of IP Spoofing :

- Unauthorized network access.
- Increased success rate of DDoS attacks.
- Difficulty in identifying the real attacker.
- Security monitoring becomes more challenging.

Prevention :

- Configure packet filtering on routers.
- Implement ingress and egress filtering.
- Use secure authentication methods.
- Monitor abnormal network traffic.

3.5 DNS Spoofing

DNS Spoofing, also known as DNS Cache Poisoning, is an attack in which false DNS records are inserted into a DNS server. As a result, users are redirected from legitimate websites to malicious websites without their knowledge.

Attackers use DNS spoofing to steal login credentials, install malware, or conduct phishing attacks.

How DNS Spoofing Works :

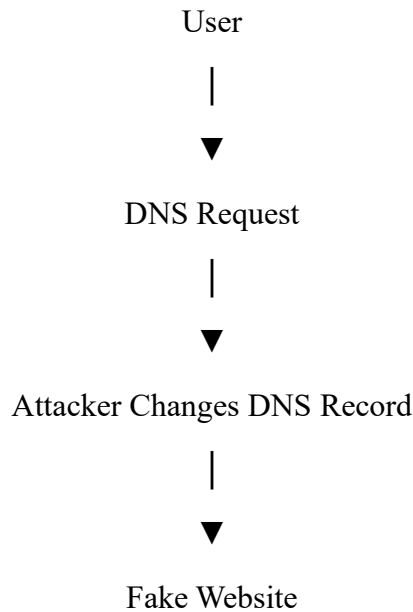
- The attacker manipulates DNS records.
- The victim enters the correct website address.
- The DNS server returns the attacker's fake IP address.
- The victim is redirected to a fake website.

Impact of DNS Spoofing :

- Phishing attacks.
- Credential theft.
- Malware installation.
- Financial fraud.
- Loss of customer trust.

Prevention :

- Use DNSSEC (Domain Name System Security Extensions).
- Regularly update DNS servers.
- Use trusted DNS providers.
- Monitor DNS traffic for suspicious activity.



3.6 Malware

Malware (Malicious Software) is any software intentionally designed to damage computer systems, steal sensitive information, or disrupt normal operations. Malware can spread through infected email attachments, malicious websites, pirated software, removable storage devices, or software vulnerabilities. Once installed, it can perform harmful activities without the user's knowledge.

Types of Malware

- Virus
- Worm
- Trojan Horse
- Spyware
- Adware
- Rootkit
- Keylogger
- Impact of Malware
- Theft of personal and financial information.
- Slow system performance.
- Data corruption or deletion.
- Unauthorized access to computer systems.
- Business disruption and financial losses.

Real-World Example

The ILOVEYOU Virus (2000) infected millions of computers worldwide through email attachments, causing billions of dollars in damages by deleting files and spreading automatically.

Prevention

- Install trusted antivirus software.
- Avoid downloading files from unknown websites.
- Keep software updated.
- Scan external devices before use.
- Avoid opening suspicious email attachments.

3.7 Ransomware

Ransomware is a type of malware that encrypts files and demands payment from the victim to restore access. Attackers usually demand payment using cryptocurrencies because they are difficult to trace.

How Ransomware Works :

- The victim downloads an infected file or clicks a malicious link.
- Malware encrypts important files.
- A ransom note appears demanding payment.
- The victim may permanently lose access if no backup exists.

Impact :

- Data loss
- Financial losses
- Business downtime
- Reputation damage
- Legal consequences

Real-World Example

The WannaCry Ransomware Attack (2017) infected more than 200,000 computers across 150 countries, affecting hospitals, businesses, and government organizations. It exploited an unpatched Windows vulnerability.

Prevention :

- Maintain regular backups.
- Install security patches promptly.
- Use antivirus software.
- Enable Multi-Factor Authentication.
- Train employees to recognize phishing emails.

4. Real-World Case Studies

Case Study 1 – Dyn DNS DDoS Attack (2016)

- In October 2016, attackers used the Mirai Botnet to launch a large Distributed Denial of Service (DDoS) attack against Dyn, a major DNS service provider. As a result, popular websites such as Twitter, Netflix, Reddit, Spotify, GitHub, and PayPal became temporarily unavailable. This incident highlighted the importance of securing Internet of Things (IoT) devices.
- Lessons Learned
- Secure IoT devices with strong passwords.
- Monitor unusual network traffic.
- Implement DDoS mitigation solutions.

Case Study 2 – WannaCry Ransomware Attack (2017)

The WannaCry ransomware attack spread rapidly across more than 150 countries by exploiting a Windows vulnerability. Hospitals, banks, universities, and businesses were affected, resulting in significant financial losses and service disruptions.

Lessons Learned :

- Regular software updates are essential.
- Backup critical data frequently.
- Patch management reduces security risks.

5. Prevention and Mitigation :

Organizations should adopt a layered security approach to minimize cyber threats.

- Recommended Security Measures
- Install firewalls and intrusion detection systems.
- Keep operating systems and applications updated.
- Use antivirus and endpoint protection software.
- Enable Multi-Factor Authentication (MFA).
- Encrypt sensitive information.
- Perform regular vulnerability assessments.
- Monitor network traffic continuously.
- Conduct regular employee cybersecurity awareness training.
- Implement strong password policies.
- Create regular data backups.

6. Best Practices

The following best practices help organizations strengthen network security:

- Keep all systems updated with security patches.
- Use secure authentication methods.
- Restrict unnecessary network services.
- Follow the Principle of Least Privilege (PoLP).
- Monitor network logs regularly.
- Educate employees about phishing attacks.
- Secure wireless networks using WPA3 encryption.
- Perform periodic security audits.
- Prepare an Incident Response Plan.
- Test backup recovery procedures regularly.

7. Conclusion

Network security has become one of the most important aspects of modern information technology. As cyberattacks continue to evolve, organizations must adopt proactive security measures to protect their networks, systems, and sensitive information. Threats such as DoS attacks, DDoS attacks, Man-in-the-Middle attacks, spoofing, malware, and ransomware can cause significant financial losses and operational disruption if not properly managed.

A combination of advanced security technologies, regular software updates, employee awareness, strong authentication, continuous monitoring, and effective security policies significantly reduces cybersecurity risks. Organizations that invest in network security are better prepared to defend against emerging cyber threats and maintain business continuity.

8. References

- NIST Cybersecurity Framework
- OWASP Foundation
- Cisco Networking Academy
- Microsoft Security Documentation
- IBM Security Learning Center
- Cloudflare Learning Center
- EC-Council Cybersecurity Resources
- Kaspersky Cybersecurity Blog